

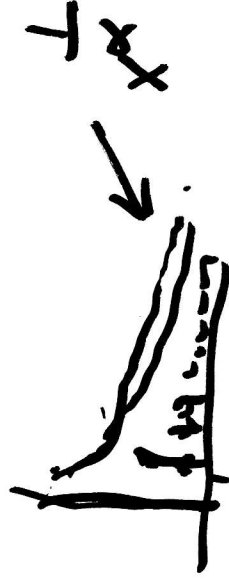
Financial Time Series

TA Session #1

3/26/21

Question #2

How Formal



Power laws

Question #3

Only assumption is ϵ_t indep
with common $E(\epsilon_t) = \mu$, $\text{Var}(\epsilon_t) = \sigma^2$

[illegible]

$$\begin{aligned} X_t &= \varepsilon_t + \varepsilon_{t-1} + \varepsilon_{t-2} \\ \text{Cov}(X_t, X_{t+h}) &= \text{Cov}(\varepsilon_t + \varepsilon_{t-1} + \varepsilon_{t-2}, \varepsilon_{t+h} + \varepsilon_{t+h-1} + \varepsilon_{t+h-2}) \\ &= \text{Cov}(\varepsilon_t, \varepsilon_{t+h}) + \text{Cov}(\varepsilon_{t-1}, \varepsilon_{t+h}) + \text{Cov}(\varepsilon_{t-2}, \varepsilon_{t+h}) \\ &\quad + \text{Cov}(\varepsilon_t, \varepsilon_{t+h-1}) + \text{Cov}(\varepsilon_{t-1}, \varepsilon_{t+h-1}) + \text{Cov}(\varepsilon_{t-2}, \varepsilon_{t+h-1}) \\ &\quad + \text{Cov}(\varepsilon_t, \varepsilon_{t+h-2}) + \text{Cov}(\varepsilon_{t-1}, \varepsilon_{t+h-2}) + \text{Cov}(\varepsilon_{t-2}, \varepsilon_{t+h-2}) \end{aligned}$$

$$x^t \parallel e^t + e^{\frac{1}{2}t} + e^{\frac{1}{2}t-2}$$

AR(1) process

$$X_t = \alpha + \beta X_{t-1} + \varepsilon_t \quad \{\varepsilon_t\} \text{ white noise } WN(0, \sigma^2)$$

$$\begin{aligned} &= \alpha + \beta [\alpha + \beta X_{t-2} + \varepsilon_{t-1}] + \varepsilon_t \\ &= \alpha(1+\beta) + \beta^2 X_{t-2} + \varepsilon_t + \beta \varepsilon_{t-1} \\ &= \alpha(1+\beta) + \beta^2 [\alpha + \beta X_{t-3} + \varepsilon_{t-2}] + \varepsilon_t + \beta \varepsilon_{t-1} \\ &= \alpha(1+\beta+\beta^2) + \beta^3 X_{t-3} + \varepsilon_t + \beta \varepsilon_{t-1} + \beta^2 \varepsilon_{t-2} \\ &= \dots \\ &= \alpha \left(\sum_{i=0}^{t-1} \beta^i \right) + \beta^t X_0 + \sum_{i=0}^{t-1} \beta^i \varepsilon_{t-i} \end{aligned}$$

if $|\beta| < 1$ $\alpha \sum_{i=0}^{t-1} \beta^i \rightarrow \frac{\alpha}{1-\beta}$ $\beta^t X_0 \rightarrow 0$ $\sum_{i=0}^{t-1} \beta^i \varepsilon_{t-i} \rightarrow \sum_{i=0}^{\infty} \beta^i \varepsilon_{t-i}$

2.2 if $|\beta| < 1$ $X_t = \frac{\alpha}{1-\beta} + \sum_{i=0}^{\infty} \beta^i \varepsilon_{t-i}$ $MA(\infty)$ $\text{Wold representation of AR(1)}$

MS

$$\text{Cov}(X_s, X_t) = \text{Cov}(X_t, X_s) = E[X_t X_s] - E[X_t] E[X_s]$$

$$\gamma(h) = \gamma(-h) \quad s < t$$

$$\text{Var}(X_t) = \text{Cov}(X_t, X_t) = E(X_t^2) - (E(X_t))^2 \quad t \geq s$$

Weak stationarity - Conditions or $\gamma(h)$
for a weak law of large numbers

$\{X_t, t \geq 1\}$ weakly stationary time series

$$EX_t = \mu, \text{Var}(X_t) = \sigma^2, \gamma(h) \text{ cov fcn.}$$

$$\frac{X_1 + \dots + X_n}{n} \longrightarrow \mu$$

Weak law $\longleftrightarrow$ Convergence
in probability

(3)

$$P(|\bar{X}_n - \mu| > \varepsilon) \rightarrow 0 \text{ as } n \rightarrow \infty \text{ for any } \varepsilon > 0$$

Convergence in probability

$$P(|\bar{X}_n - \mu| > \varepsilon) \leq \frac{\text{Var}(\bar{X}_n)}{\varepsilon^2} \xrightarrow[\text{as } n \rightarrow \infty]{\text{Chebyshev's Ineq}} 0$$

$$= \frac{\text{Var}(X_1 + \dots + X_n)}{n^2 \varepsilon^2} \quad (\text{if indep}) \quad \xrightarrow[\text{as } n \rightarrow \infty]{\text{Chebyshev's Ineq}} 0$$

$$= \frac{n\sigma^2 + \text{covariance terms}}{n^2 \varepsilon^2}$$

$$= \frac{\text{Cov}(X_1 + \dots + X_n, X_1 + \dots + X_n)}{n^2 \varepsilon^2} = \frac{(1 \cdot n)\sigma^2 + \dots + (1 \cdot n)\sigma^2 + n\sigma^2}{n^2 \varepsilon^2}$$

$$= n\sigma^2 + 2 \sum_{i=1}^{n-1} (n-i) x(i) \quad \xrightarrow{0 \leftarrow \text{as } n \rightarrow \infty} \frac{23n}{2}$$

$$(1) \quad \underline{\underline{\sum_{i=1}^{\infty} |x(i)| < \infty}}$$

$$\text{Bound numerator} \quad \frac{n\sigma^2 + 2n \sum_{i=1}^{n-1} |x(i)|}{n^2}$$

$$(2) \quad \underline{\underline{\sum_{i=1}^{n-1} (n-i) x(i) \rightarrow 0 \quad \text{as } n \rightarrow \infty}}}$$

$$\frac{n\sigma^2 + 2n \sum_{i=1}^{\infty} |x(i)|}{n^2} \rightarrow 0$$

$$\text{or } \underline{\underline{\sum_{i=1}^n |x(i)| \rightarrow 0 \quad \text{as } n \rightarrow \infty}}}$$

(3)

$$\text{AR}(1) \quad x(i) \approx \frac{1}{2} B^i$$

$$\sum |B|^i < \infty$$

Returns over 1 day
Returns over 30 days

$$\begin{array}{ccccccc} x_2 & x_3 & x_4 & \dots & x_n & & \\ x_1 & x_2 & x_3 & & x_{n-1} & & \\ x_{30}/x_1 & x_{60}/x_{31} & x_{90}/x_{61} & \dots & x_n/x_{n-1} & & \end{array}$$

$$\frac{n-1}{2500}$$

$$\frac{n-1}{30}$$

$$\left[\frac{1}{11/2011}, \frac{3}{20/21} \right] \leftarrow$$

$$\left[\frac{1}{1/2000}, \frac{14}{31/2019} \right] \leftarrow$$

zoom

$$\left[2018 \right]$$